Ferrari
Ferrari

Vehicle platform

Of ECUs, microcontrollers and simulators…

Course outline

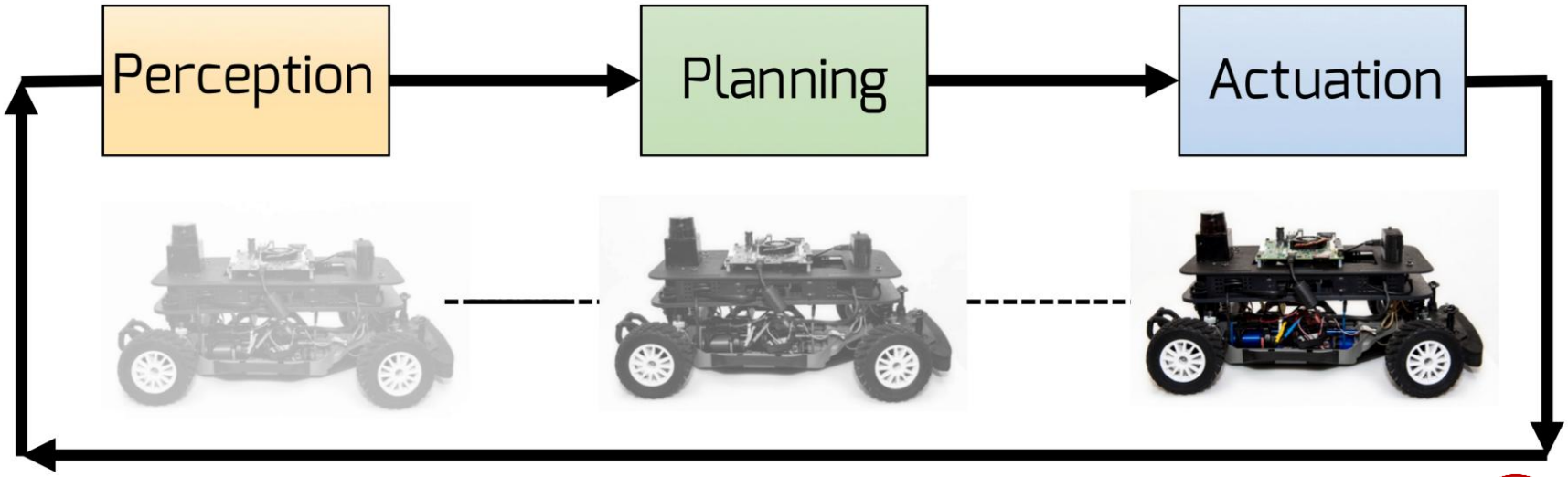
Lesson #1

- Intro course + basics of AD
- Perception and AI
- Hardware platforms

Lesson #2

- Cars and the city
- Navigation: planning and control
- Let's build the car

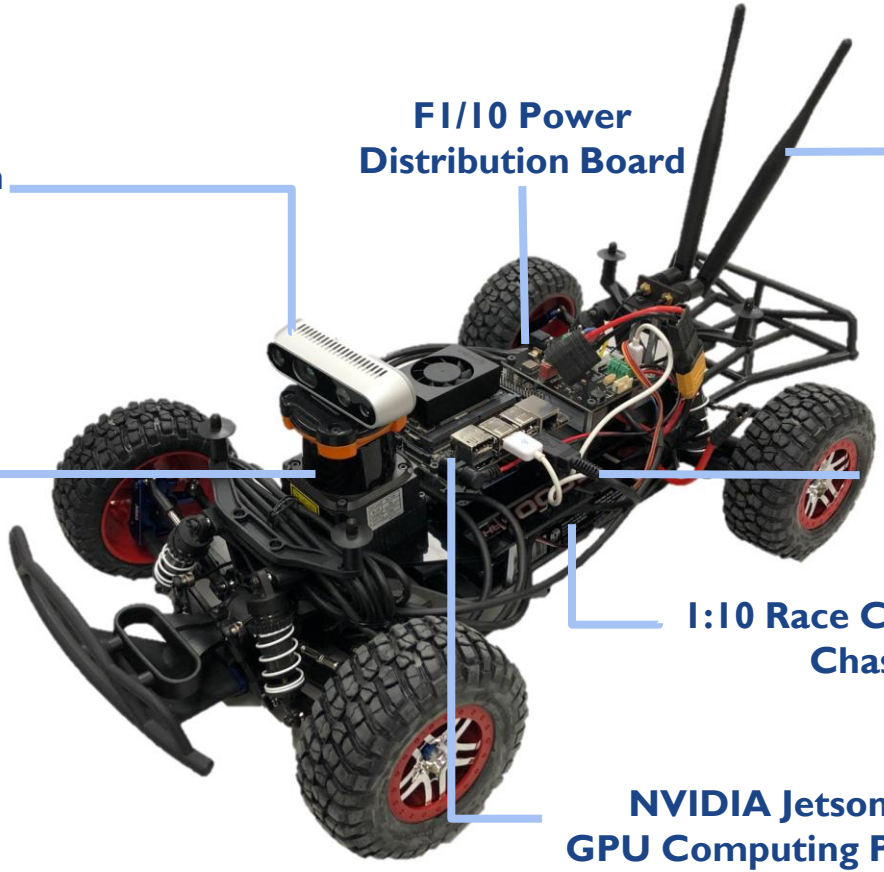
Assignments



Real-Time constraint!

(At least) three stages – F1/10

The F1/10 platform



Stereovision Camera

F1/10 Power Distribution Board

WiFi Control

Lidar

Single 5000 mAh LiPo Battery

1:10 Race Car Vehicle Chassis

NVIDIA Jetson NX GPU Computing Platform

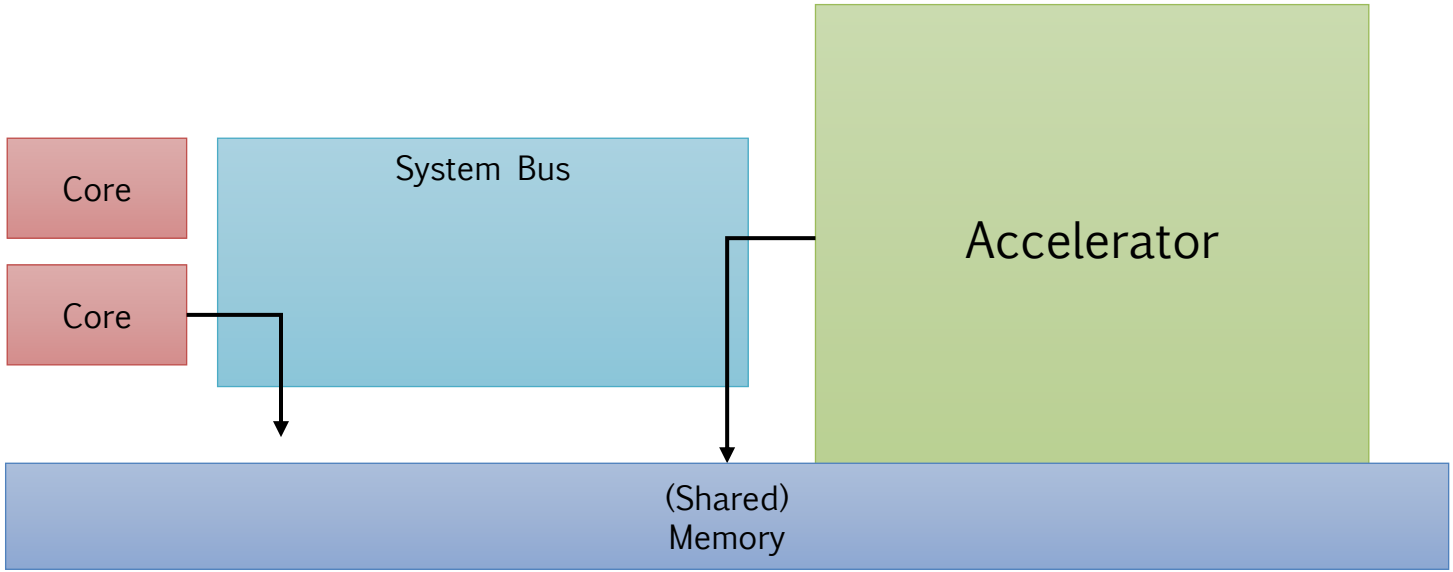
F1 TENTH

On-board computer - template

Aka: Domain controller, aka: ECU, aka: Zonal controller, aka …

Embedded heterogeneous platform

- Multi-core + data cruncher accelerator
- Shall employ safety core (Aurix?) to enable automotive ISO26262/ASIL-D compliancy



NVIDIA Jetson Orin

6/8 core ARM host

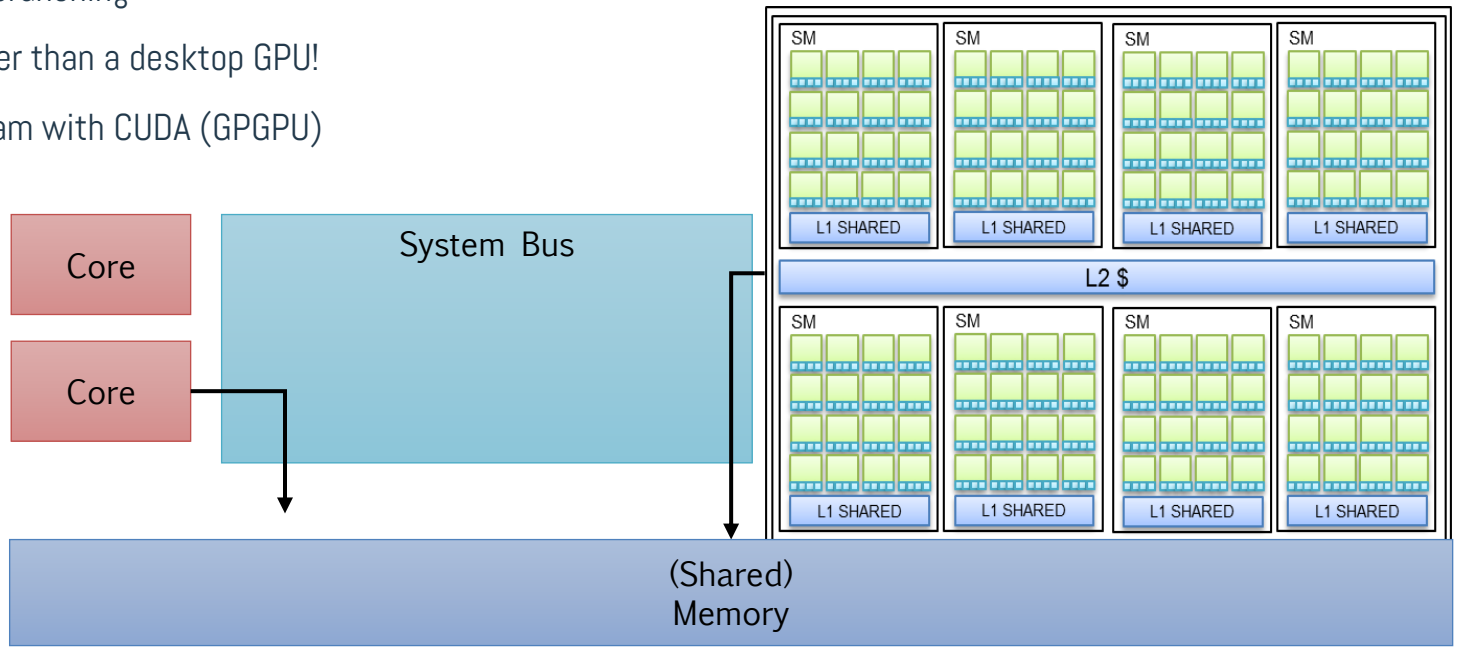
› For control-based workload

Embedded GPU

› Data crunching

› Smaller than a desktop GPU!

› Program with CUDA (GPGPU)



Xilinx UC+ / Kria

6 core ARM host

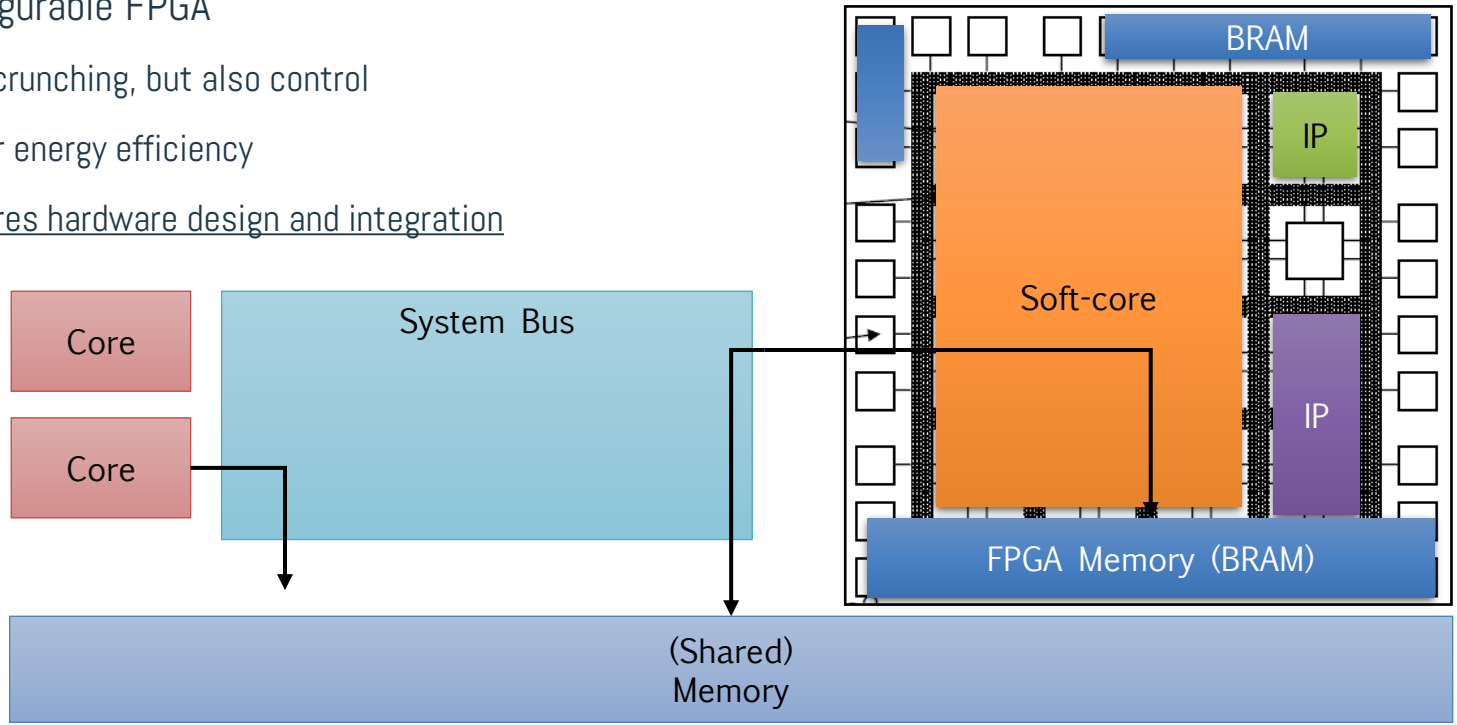
› For control-based workload

Reconfigurable FPGA

› Data crunching, but also control

› Higher energy efficiency

› Requires hardware design and integration



Our F1/10 cars

With GPGPUs

› Bud => NVIDIA Orin

› Kitt Jr. => NVIDIA Xavier NX

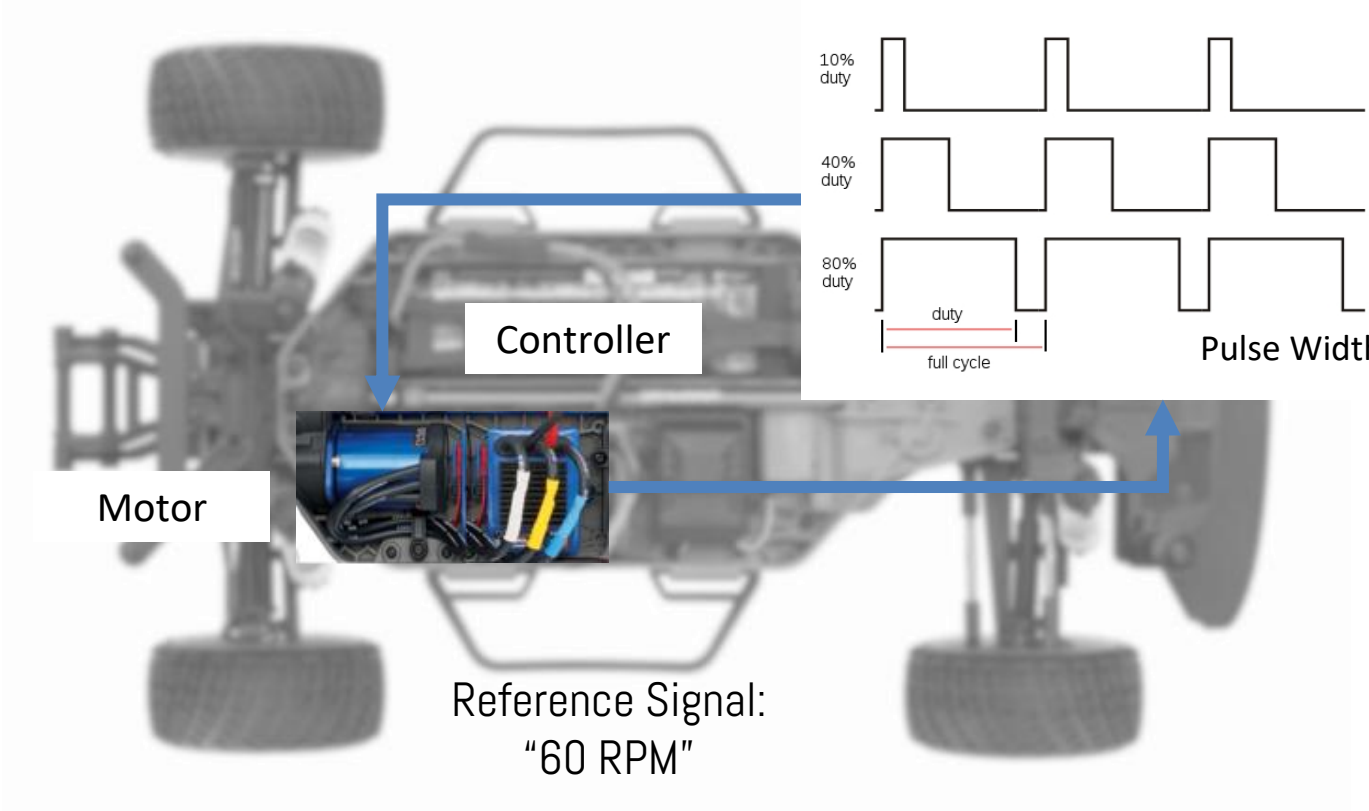
With FPGA

› Thundershot F1/10 => Xilinx Kria

› Frankenstein/Frankie
  - Whatever we want to test
  - Currently, cybersecurity

Actuation circuit

Remember: every vehicle is somehow unique!



Pulse Width-Modulated signals

PWM - D/A conversion (recap..?)

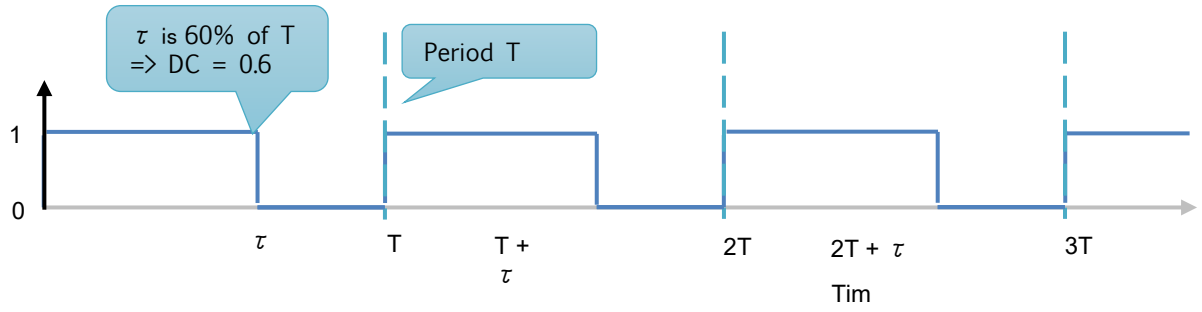
Generate a tension corresponding to a digital value stored in a register

› Not easy! Use Pulse-Width Modulation (PWM)
  – (Almost) fully implementable in SW!

How it works

DC = τ / T

1. Generate a periodic signal of amplitude 1 whose **duty cycle** is proportional to the digital value we want to convert
2. Give it to a low-pass filter, to average (such as Resistor-Capacitor - RC circuit)
3. ..and enjoy your analog signal! ☺

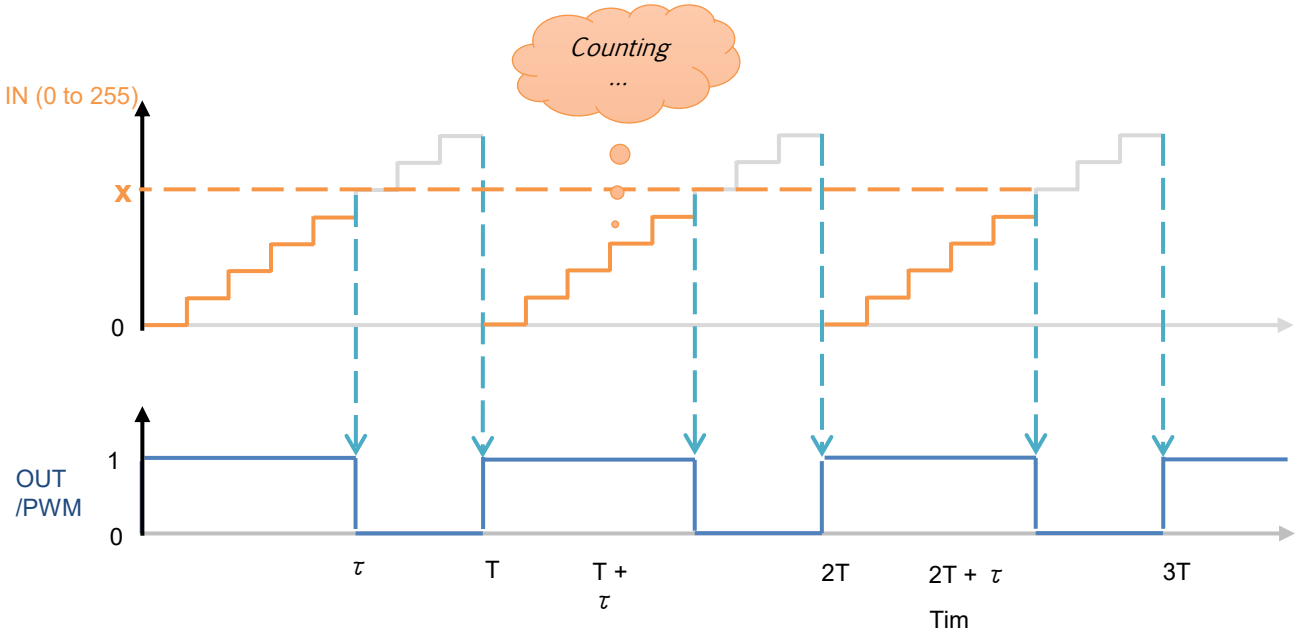


PWM (recap..?) – step 1

We need

› A register that counts from 0 to 255 (8-bit)

› An output port (bit) set to '1' and becoming '0' when input value matches the one of the register

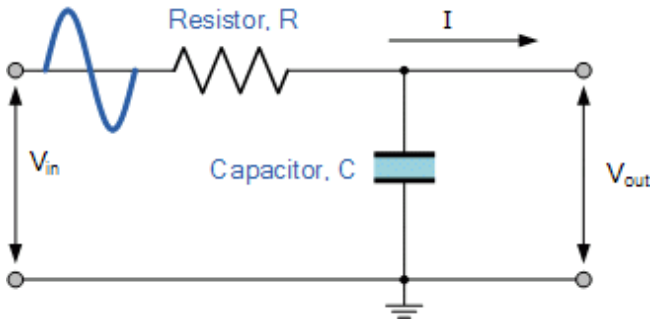


RC low-pass filter (recap..?)

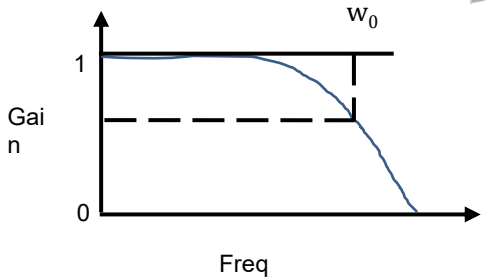
Simple electric circuit with a Capacitor and a Resistance

› "Averages" the IN value

$$|V_{out}| = |V_{in}| \times \frac{1}{\sqrt{1 + w^2 R^2 C^2}}$$



Frequency domain



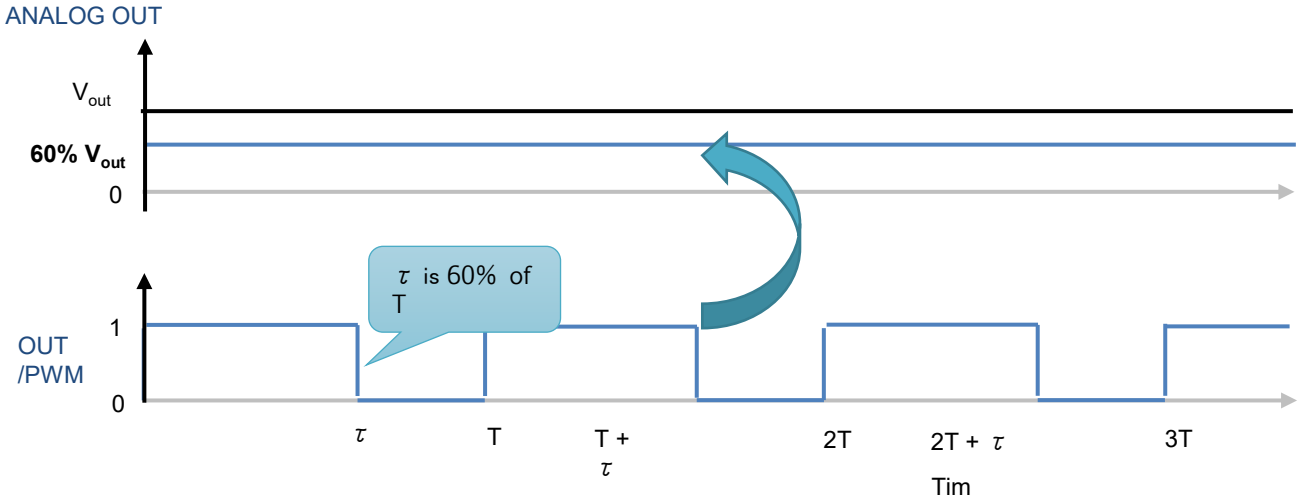
Cutoff freq $w_0 = \frac{1}{RC}$

PWM (recap..?) – step 2

Now, compute the average for every T

- Using a low-pass filter
- Plug it to output port
- *Et voilà*

Extremely useful in engine controls



Electronic Speed Controller – (V)ESC

Manages the brushless engines of the car

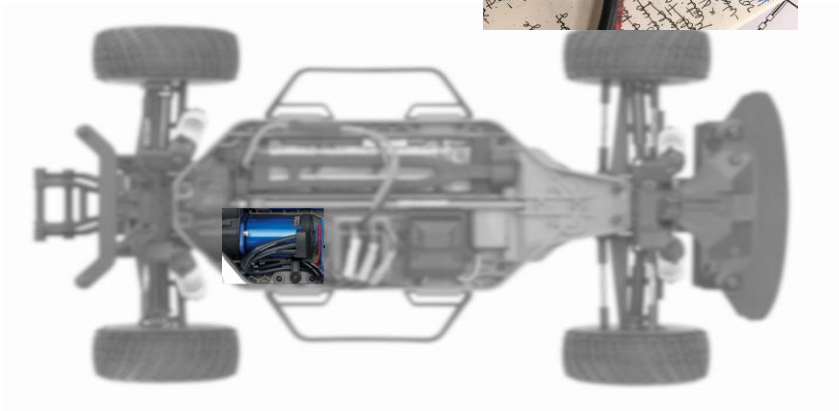
https://vesc-project.com/

Programmed via a tool

› Once-for-all

Accepts commands for lat/lng control

› Acceleration & steering



Actuation system W/VESC

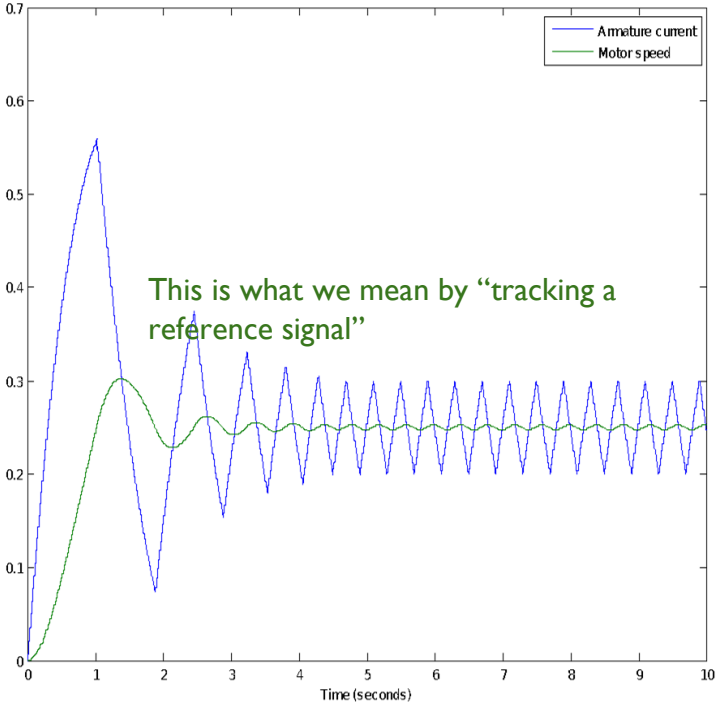
Reference
r(t)
("60rpm")

CONTROLLER

ENGINE

Output *x(t)*
(angular velocity)

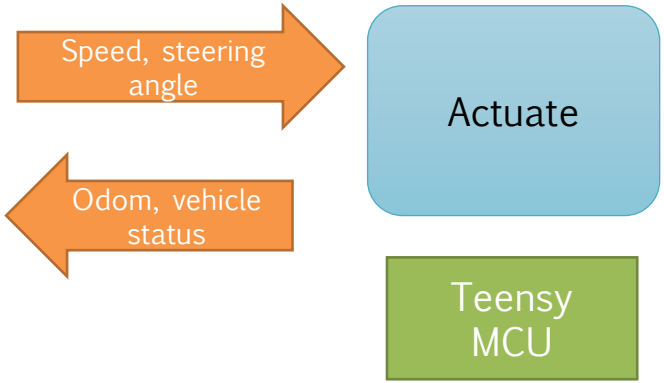
Control input
u(t)
(electric current)

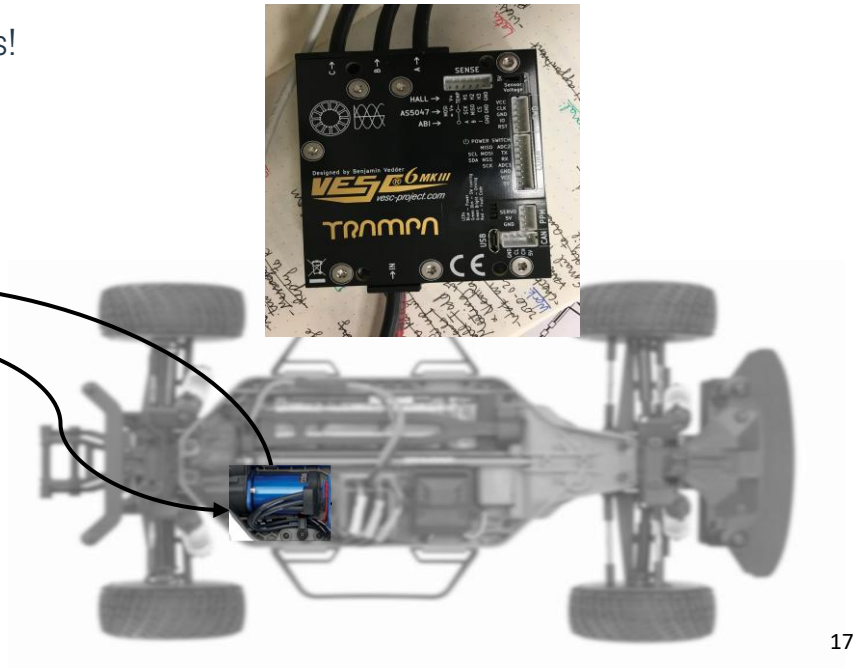


«Old» actuation based on Teensy μcontroller

VESC accessed via UART

- The navigation ECU had only CAN/serial port
- Teensy converted the signal – teensy has no OS
- Typical scenario, also real cars have legacy actuator ECUs!

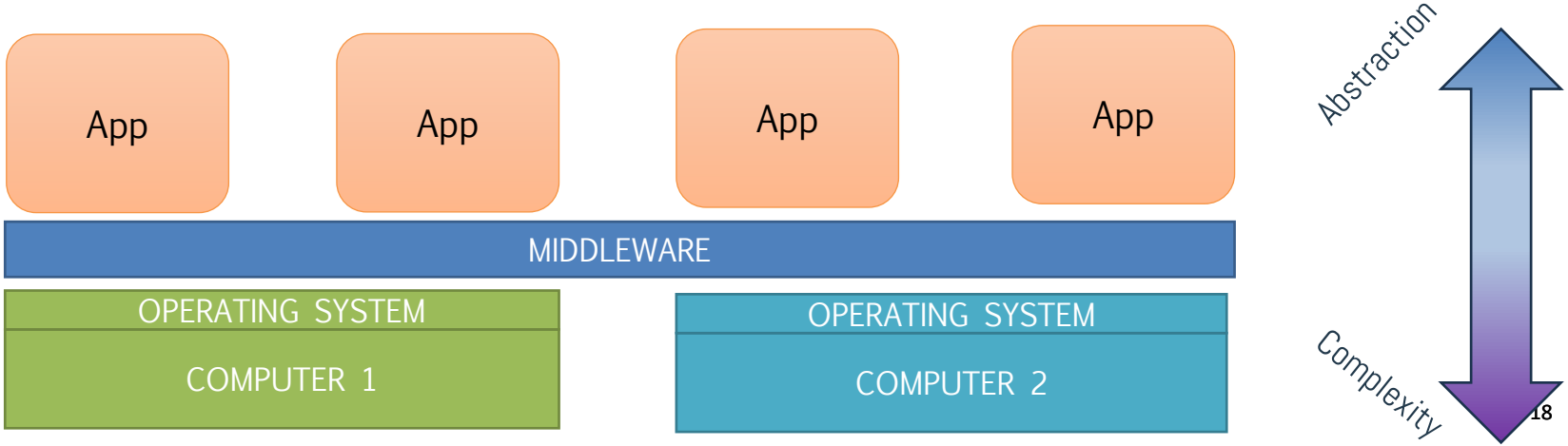




A quick travel…in programmer's mind

Computer science is all about hiding hardware

- We typically represent it at the bottom of figures
- At the top you have applications
- In the middle, other components each of which has a specific responsibility



As applied to our AD stack

Robot Operating System (ROS)

- A middleware that hides the actual computer and OS we run on
- Application components are (wrapped) called ROS **Nodes** that exchange ROS **Messages**
- Most of sensors already publish ROS messages!

Actuation timings

- LiDAR publishes @40Hz (25ms)
- Teensy loop @?? Hz
  - We don't care! If we don't send a signal, it holds the previous one
- Our pipeline must meet 25ms deadline…incl. ROS delays!

Point cloud

ROS

Urg

Delay

ROS

Perc/Loc

Delay

ROS

Plan+Ctrl

Delay

ROS

Actuate

UART

VESC (out of our control)

ROS2 Middleware

GNU/Linux

NVIDIA Jetson NX

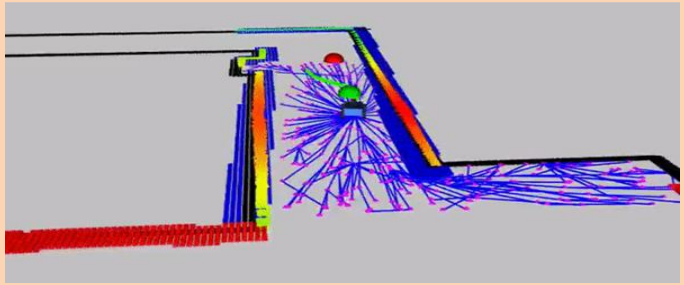
µkernel

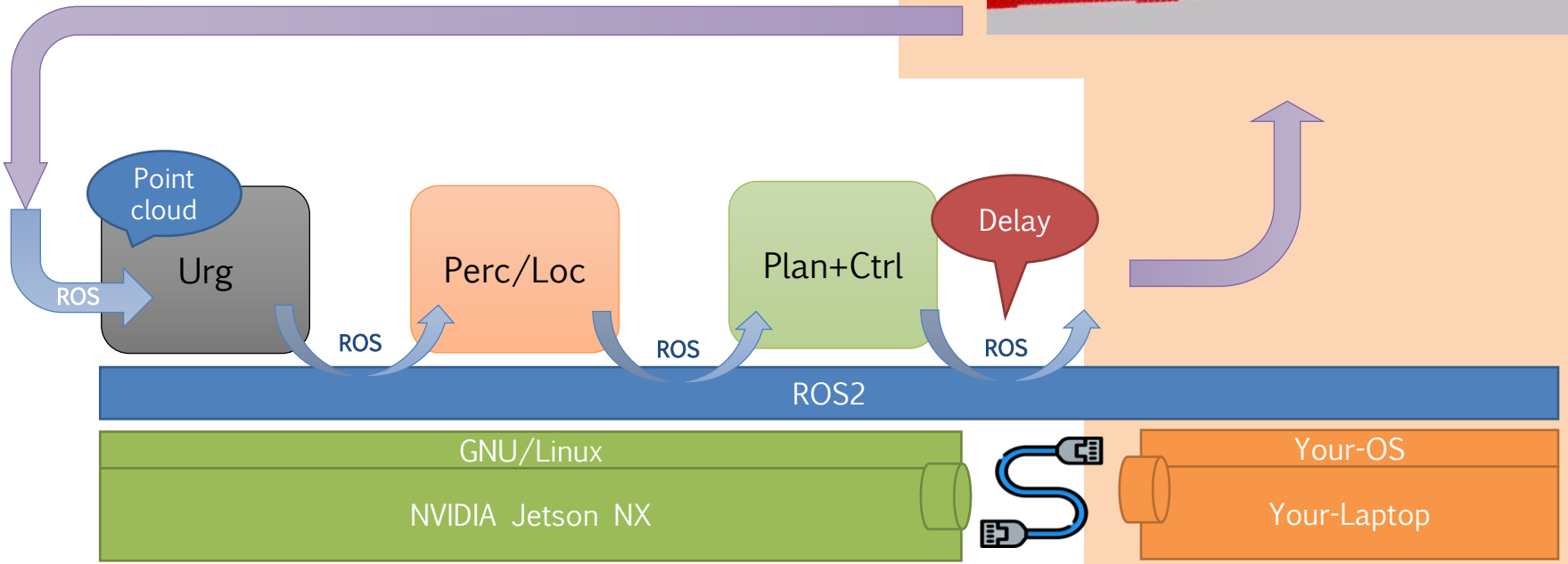
Teensy

Interaction with a simulator

Most of simulators have can exchange ROS messages!

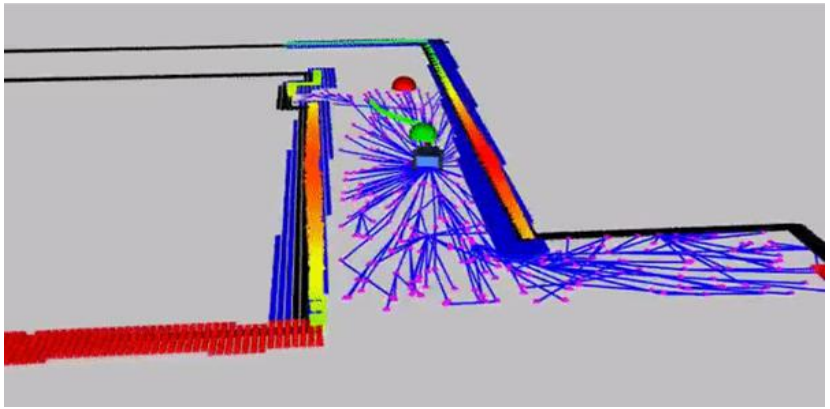
- Can receive commands
- Send data from *sumulated* sensors
- ROS ensures that every exchange is machine and OS independant





F1/10 Gym

- Lightweight 2D simulator built in Python
- Faster than real-time execution (30x realtime)
- Runs multiple vehicle instances
- Publishes laser scan and odometry data
- Built for fast prototyping



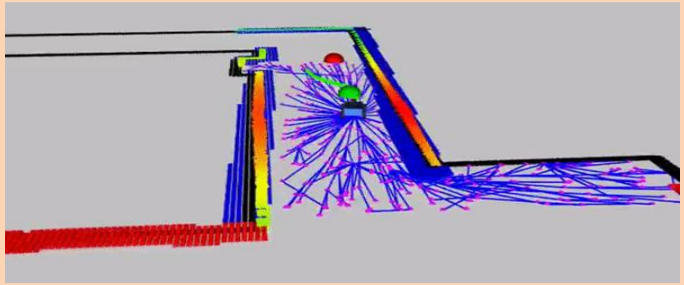
Where to run application components?

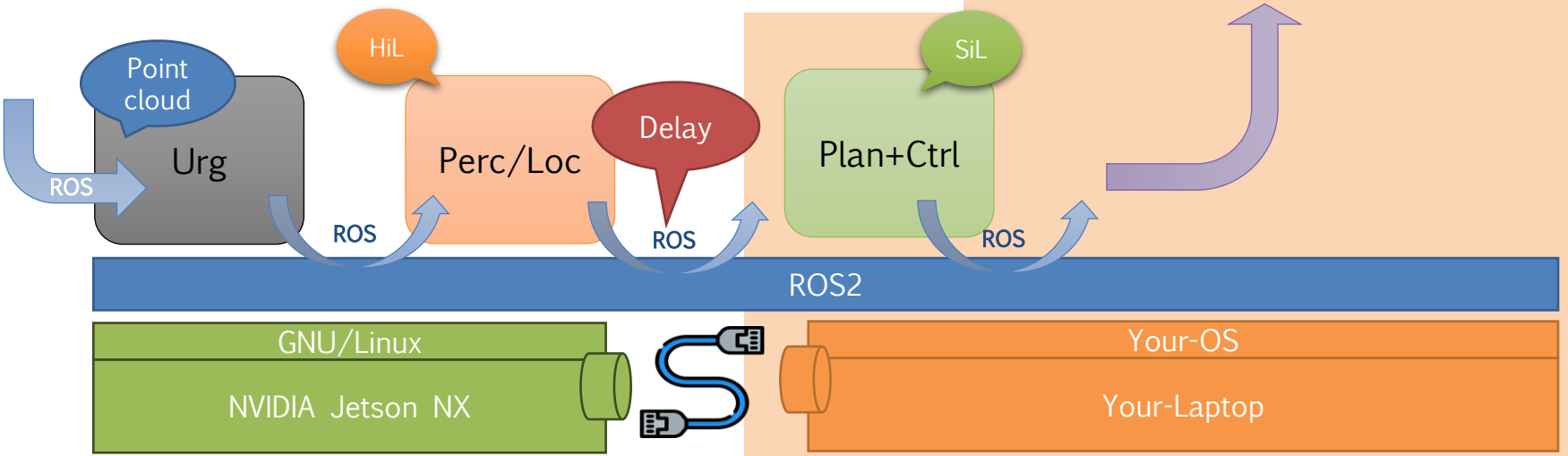
Hardware-in-the-loop - HiL

- Run them in the very same ECU hardware
- With delays due to comm, but execution is the same than on real ECU

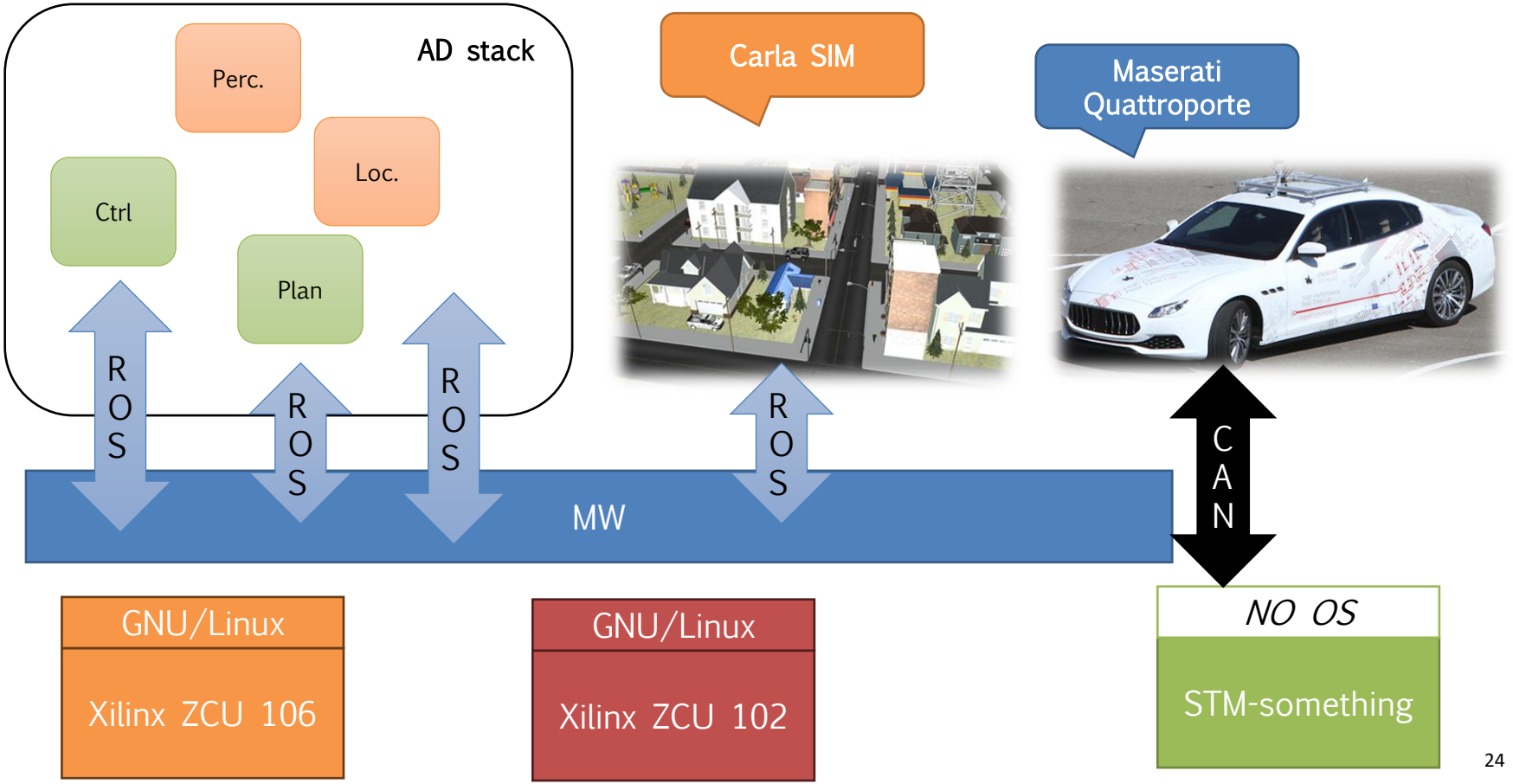
Software-in-the-loop - SiL

- Run them in the same computer where the simulator resides

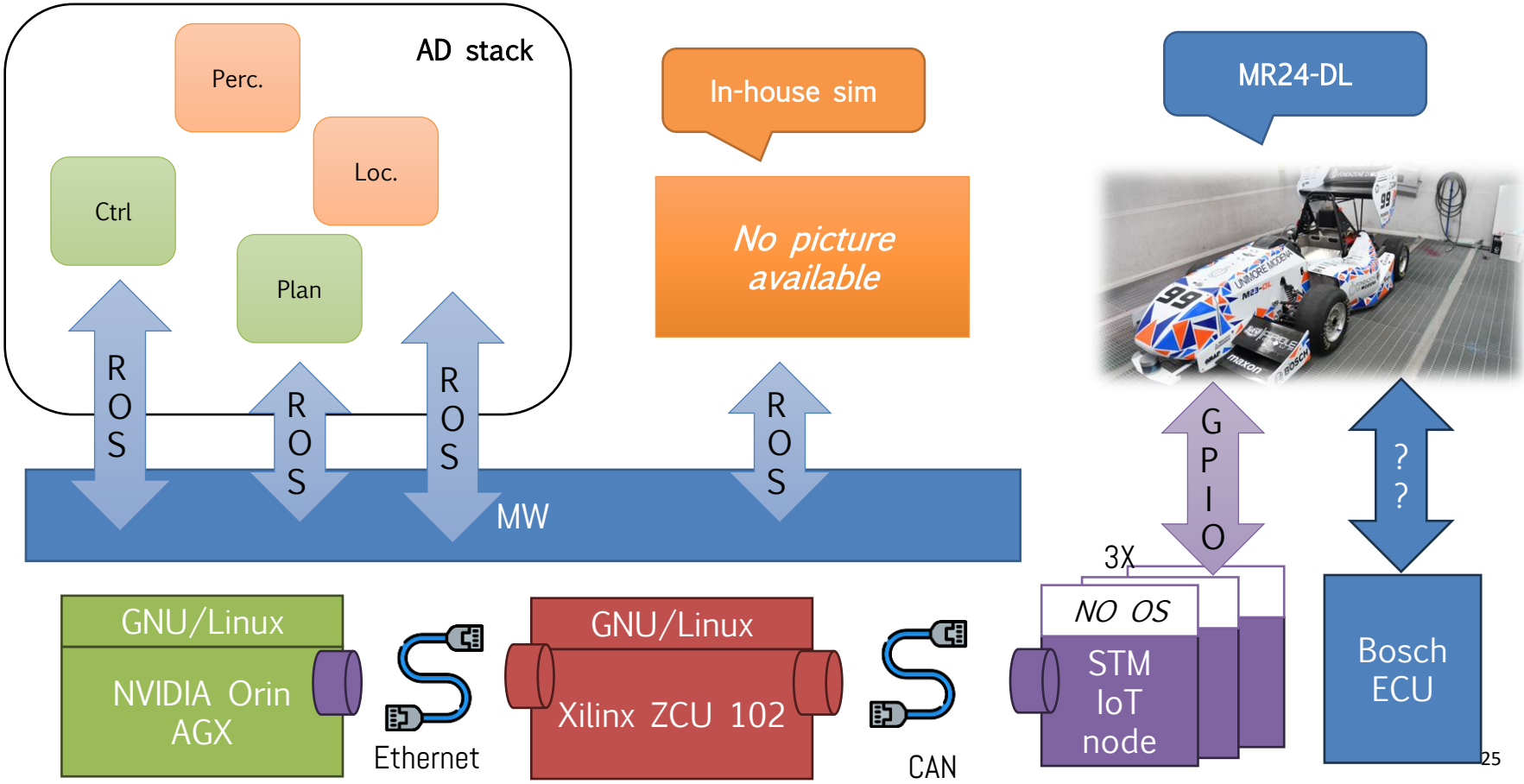




The Thundershot project – A more complex example



Formula Student



Assignments

I can send you a quiz which you need to solve and send back to me

› By friday

..or…you can challenge yourself in one of the following (with support by me and the guys from the lab)

› Building a simple node for AD components (e.g., pure pursuit, follow-the-gap) – Python or C++

› Implement some model of a vehicle (Matlab/Simulink)

› Your ideas…?

No deadline for the latter, the earlier the better

References

Course website

- https://github.com/HiPeRT/F1tenth-PhD

My contacts

- paolo.burgio@unimore.it
- http://hipert.mat.unimore.it/people/paolob/

Resources

- https://f1tenth.org
- A "small blog"
  - http://www.google.com